Do Now · retrieval

Exam-style data representation problems

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does CountAtLeast([5,7,3],5) return?

Q2

What does O(log n) often describe in binary search?

Q3

What is the sign bit for a negative 8-bit two's-complement integer?

Q4

Calculate $0.1000_2 \times 2^3$.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Respect the bit width — Write positive 5 using exactly eight bits.

Core idea 2

Invert then add one — The result is 11111011_2, representing −5 in 8-bit two's complement.

Core idea 3

Interpret floating-point differently — 0.1100_2 is fractional 0.75; multiplying by 2^3 gives 6.

Key visual · complete concept

Exam-style data representation problems

+5	**00000101**	**Positive**
INVERT	**11111010**	**All bits**
ADD 1	**11111011**	**Represents −5**

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

bit width — Number of bits in a representation.

two's complement — Fixed-width signed binary format.

mantissa — Signed fractional significant component.

exponent — Power-of-two scaling value.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Convert −10 to 8-bit two's complement.

Follow the worked method

1. Predict — Write 00001010 for +10 and invert to 11110101.
2. Explain — Add one: 11110110 represents −10.

Source model · final illustrated step

Convert −10 to 8-bit two's complement.

+10	**00001010**	**Start**
INVERT	**11110101**	**All bits**
ADD 1	**11110110**	**−10**

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

Start with +7 = 00000111 and invert all bits.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Convert unsigned binary 00001011 to decimal.

Question 2

Encode −12 in 8-bit two's complement.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Normalise positive $0.0011_2 \times 2^5$ using a two's-complement fractional mantissa.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Eight-bit two's complement can represent decimal +128.

Use key vocabulary from memory

1. Explain bit width in your own words.

2. Explain two's complement in your own words.

3. Explain mantissa in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.